

Qatar-9b

R-2208

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_260530 · created 2026-08-14T01:23:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461190.69 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2 +/- 0.06
Transit depth (Rp/Rs)^2	4.0 +/- 2.39 [%]
Orbital Inclination (inc)	88.69 +/- 1.63
Airmass coefficient 1 (a1)	0.034 +/- 0.021
Airmass coefficient 2 (a2)	2.22 +/- 0.85
Scatter in the residuals of the lightcurve fit is	62.32 %
Best Comparison Star	#3 - [97, 342]
Optimal Aperture	0.87
Optimal Annulus	3.0
Transit Duration (day)	0.064 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.2 ± 0.06 vs 0.1489 (deviation 0.81σ)
Duration fit vs published	0.064 d vs 0.0779 d (deviation 1.2σ)
Mid-time O–C	32.1 min
Depth SNR	3.2
Residual scatter	62.32 %

Light curve

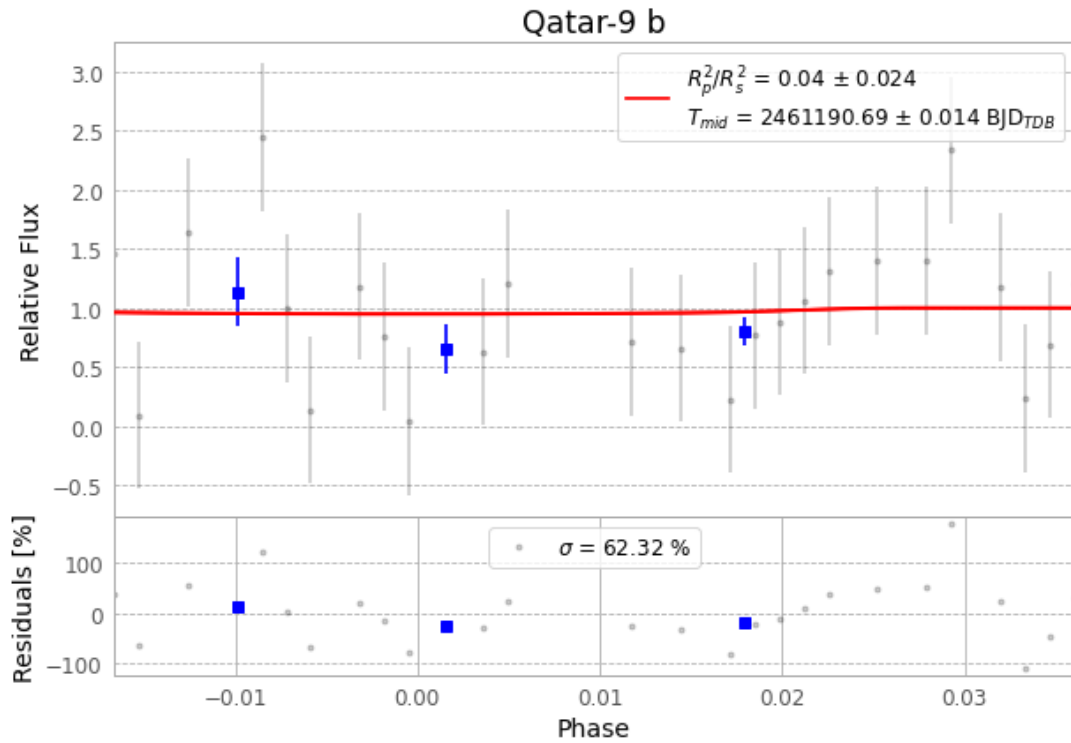


Figure 1 — FinalLightCurve_Qatar-9 b_2026-05-29.png (SHA-256 56dac940fa4a6136...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [516, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f1fe389948c7a7b7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

40 FITS frames (26 MB), Qatar-9260530035715.FITS → Qatar-9260530055416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-260530.log (d80d20529f26...), FinalLightCurve_Qatar-9 b_2026-05-29.png (56dac940fa4a...), FinalLightCurve_Qatar-9 b_2026-05-29.csv (318ca6938f08...)

Compute resources

Wall time 92.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 01:23Z.